

Food chains

An ocean food chain

Near the surface of the ocean, where bright sunlight strikes the water, billions of microscopic algae photosynthesize to make food. In doing so, they kick-start a food chain that ends with some of the biggest meat-eaters on the planet.

Living things rely on one another for nourishment. Energy in a food chain travels from the sun to plants, then animals, and finally to predators at the very top of the chain.

The sun provides the ultimate source of energy for life on Earth. Plants and algae change its light energy into chemical energy when they photosynthesize. Vegetarian (herbivorous) animals consume this food and they, in turn, are eaten by meat-eating carnivores. Energy is passed up the chain, and also transfers to scavengers and decomposers (see pp.146–147) when they feed on the dead remains of organisms.

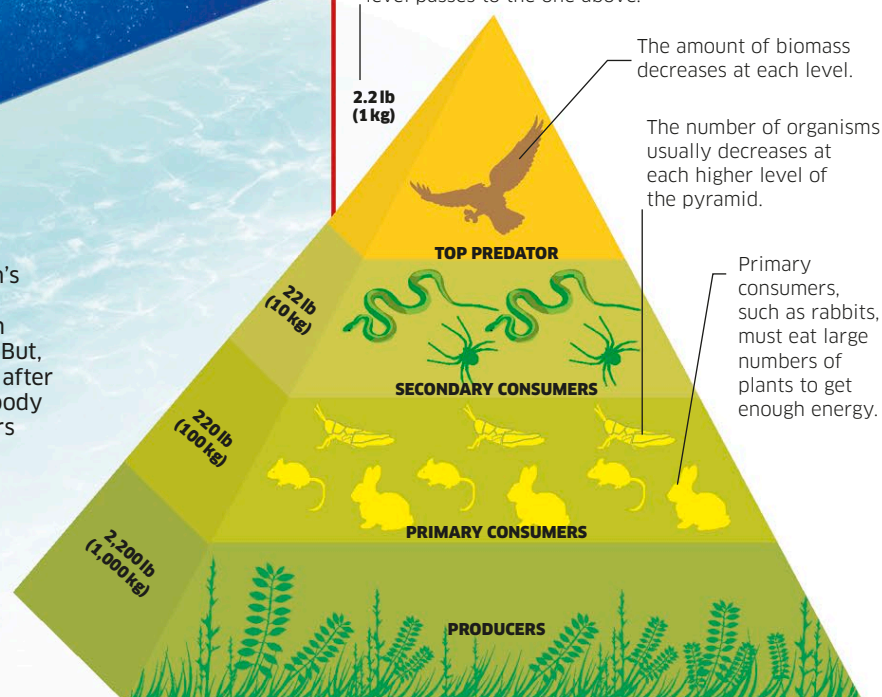
Heat production

The chemical reactions that take place in living organisms generate heat, which escapes into the surrounding water.

Ecological pyramids

The levels of a food chain can be shown stacked up together to make an ecological pyramid. Plants or algae—the producers of food—form the base of the pyramid, with consumers on the higher levels. Each stage of the pyramid can also be shown as the total weight of the organisms on that level—their biomass. Both biomass and, usually, the number of animals decreases toward the top, as energy is lost at each level. Organisms use energy to stay alive and it is given off as waste and heat, leaving less to be passed on.

Only about 10 percent of the energy, and biomass, in any level passes to the one above.



6 Great white shark Being the food chain's top predator means that little else will prey on an adult great white shark. But, like all other organisms, after death the energy in its body will support decomposers that feed on its corpse.

5 Sea lion Sea lions swim hundreds of yards from the shoreline to reach the best fishing grounds. As they hunt herring, the energy in the fish meat passes into the sea lion's body. Because their herring prey are also meat-eaters, this makes sea lions tertiary consumers.